QTM 220: Lecture 2

Population Inference for Proportions

Review

- Last lecture, we discussed description/summarization with means, conditional means, and differences in conditional means.
- We considered the properties of these summaries (e.g., balance point, least squares, etc.) and how much was left out by summarizing (e.g., squared residuals)
- This was done in the context where we only cared about the sample, either provisionally or because we had sampled everyone (i.e., a census).

Today

- Today we consider the context where we have a random sample from a large population and we have measured a binary outcome.
- Our goal over the next couple of lectures is to summarize the population with population means, conditional population means, and differences in conditional population means.
- Summarizing data we don't have (everything in the population that isn't in the sample) is more difficult than summarizing data we do have (sample).

Sample

Suppose that a week before the 2020 presidential election, you contacted a sample of $n = 625$ potential Georgia voters, randomly selected (with replacement) from the population of eligible voters $N = 7,233,584$ on the public voters register, to ask whether they planned to vote.

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Suppose also that all potential voters can be contacted, will respond honestly to your questions, and will not change their minds about voting.

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Sample

i	1	2	3	4	...	625	$\bar{y}_{625}$
y_i	1	1	0	1	...	0	.68

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Remember that the mean of a binary variable is a proportion.

Sample Versus Population

Our estimate of voter turnout from this sample is the proportion of individuals who said they would vote in the upcoming election (1s), which is the mean of Y

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We have the opportunity to see whether our sample proportion was "right" when, one week later, the election occurs and we can see the true population proportion– how many eligible voters actually turned out to vote.

Sample Versus Population

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After election day, we found that in fact $4,999,960$ people, or roughly 70% of registered voters, turned out to vote. Each eligible voter has a unique voter ID number that we can use to index the population. For the purposes of this example, let the voter ID numbers be $[1, 7233584]$

Population

ID	1	2	3	4	...	...	...	...	7,233,584	$\bar{y}_{7,233,584}$
y_{ID}	0	1	0	1	...	...	...	...	1	.70

Two Questions

1. .68 is close to .70, did we get lucky?
2. Could we have predicted how close we would get before seeing the .70?

Sampling Distribution of $\overline{Y}_{625}$

How did we draw our sample of 625 eligible voters? We randomly drew 625 voter IDs from $[1, 7233584]$ with replacement

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i	1		2		...	625		
	ID_1	Y_1	ID_2	Y_2	...	ID_{625}	Y_{625}	$\overline{Y}_{625}$
	9562350	1	8763351	1	...	1294801	0	.68

Sampling Distribution of $\overline{Y}_{625}$

i	1		2		...	625		
s	ID_1	Y_1	ID_2	Y_2	...	ID_{625}	Y_{625}	$\overline{Y}_{625}$
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70

Sampling Distribution of $\overline{Y}_{625}$

i	1		2		...	625		
s	ID_1	Y_1	ID_2	Y_2	...	ID_{625}	Y_{625}	$\overline{Y}_{625}$
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70
3	5129936	0	10981600	0	...	4096184	1	.75
4	803605	0	7036389	1	...	803605	0	.73
5	148567	0	3833847	1	...	4769869	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	4163458	0	8384613	1	...	377981	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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$f_{\cdot}(\cdot)$								

- What is the distribution of Y_1?

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$f_{\cdot}(\cdot)$		Bern(.70)		Bern(.70)			Bern(.70)	

- What is the distribution of Y_1?
- What is the distribution of $\sum_i^{625} Y_i$?

Sampling Distribution of $\overline{Y}_{625}$

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s	ID_1	Y_1	ID_2	Y_2	...	ID_{625}	Y_{625}	$\overline{Y}_{625}$
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⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
$f_{\cdot}(\cdot)$	Bern(.70)		Bern(.70)			Bern(.70)		$\frac{Bin(625,.70)}{625}$

- What is the distribution of Y_1?
- What is the distribution of $\sum_i^{625} Y_i$?
- What two numbers show that we sampled with replacement?

Review of Bernoulli and Binomial Distributions

We can write distributions for our random variables (RVs) that describe the probability of getting particular values for these variables.

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$Y_1, ..., Y_{625}$ are Bernoulli RVs:

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f_{Y_1}(y_1) &= (.71)^{y_1}(.29)^{1-y_1} \text{ for } y_1 = 0, 1 \\
\vdots \quad &= \qquad\qquad \vdots \\
f_{Y_{625}}(y_{625}) &= (.71)^{y_{625}}(.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1
\end{aligned}
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f_{Y_{625}}(y_{625}) &= (.71)^{y_{625}}(.29)^{1-y_{625}} \text{ for } y_{625} = 0, 1
\end{aligned}
$$

$\sum_{i=1}^{625} Y_i$ is a Binomial RV:

$$
f_{\sum_{i=1}^{625} Y_i}(\sum_{i=1}^{625} y_i) = \binom{625}{k}(.70)^{\sum_{i=1}^{625} y_i}(.30)^{625-\sum_{i=1}^{625} y_i} \text{ for } \sum_{i=1}^{625} y_i = 0, \ldots, 625
$$

The Sampling Distribution in R

Resample the number of voters 1,000,000 times and store these 1,000,000 numbers in a vector.

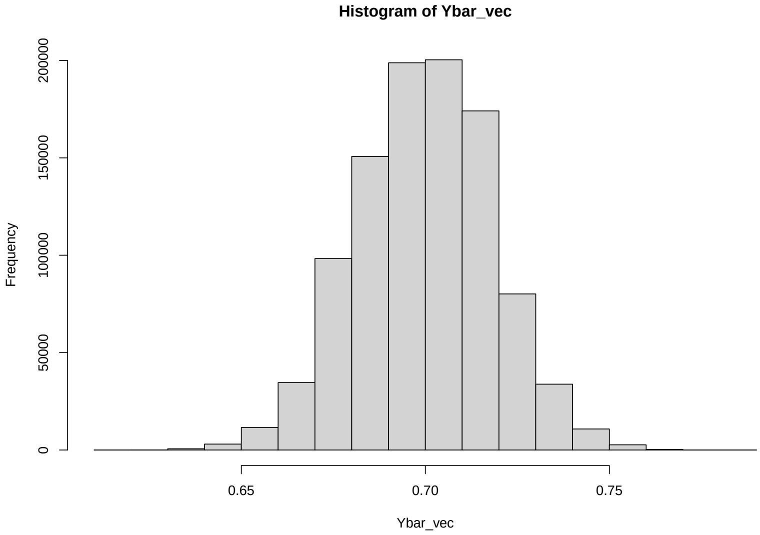
```
sumY_vec <- rbinom(1000000, size=625, prob=.70)
```

Divide all of these numbers by the sample size

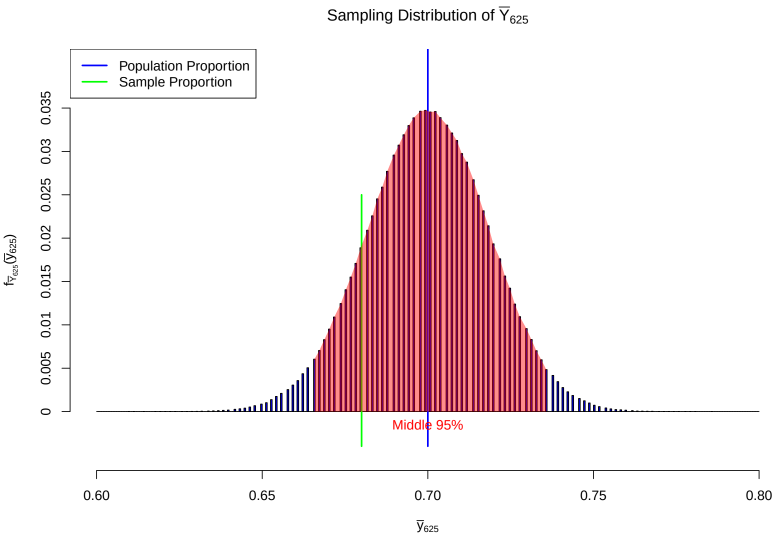
```
Ybar_vec <- sumY_vec/625
```

The Sampling Distribution Histogram in R

```r
hist(Ybar_vec)
```



A Better Sampling Distribution Histogram in R



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2. Could we have predicted how close we would get before seeing the .70?

Predicting before election day

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Predicting before election day

Could we have predicted how close we would get? Yes, in a sense.

- We will use a confidence interval.
- A confidence interval has a width, which answers the "how close" question.
- However, a confidence interval also has a coverage probability. This probability will almost never be 100%.

CI Coverage of $\overline{Y}_{625} \pm .01$

As an exercise, let's arbitrarily choose a width of 2% and use the sampling distribution.

i	1	2	...	625		
	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	$\overline{Y}_{625} \pm .01$
	1	1	...	0	.68	[.67,.69]

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2	0	1	...	1	.70	[.69,.71]
3	0	0	...	1	.75	[.74,.76]
4	0	1	...	0	.73	[.72,.74]
5	0	1	...	1	.69	[.68,.70]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
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Calculate coverage

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mean(Ybar_vec-.01 <= .7 & Ybar_vec+.01 >= .7 )
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[1] 0.399883
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```

```
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```

about 40%

Calibrating the CI

The previous exercise doesn't address our question because

1. We won't have access to the sampling distribution before election day.
2. We don't want to choose the width and receive the coverage, we want to choose coverage and receive the width

There are many ways to calibrate a CI (choose a coverage, receive the width) without observing the sampling distribution. We will explore the percentile bootstrapping approach for getting a CI with 95% coverage for the remainder of this lecture.

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However, we can take repeated random samples *with replacement* of size 625 from the sample of size 625.

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The is equivalent to replacing .70 with .68 in the R code.

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```
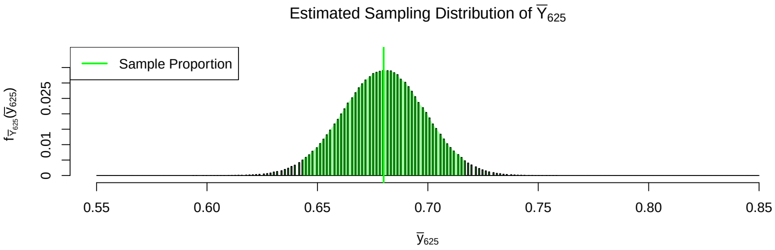
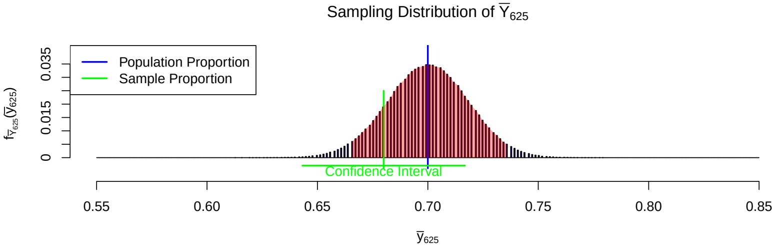



Estimated Sampling Distribution of $\overline{Y}_{625}$
Sample Proportion
$f_{\overline{Y}_{625}}(\overline{y}_{625})$
0
0.01
0.025
0.55
0.60
0.65
0.70
0.75
0.80
0.85
$\overline{y}_{625}$



Sampling Distribution of $\overline{Y}_{625}$
Population Proportion
Sample Proportion
$f_{\overline{Y}_{625}}(\overline{y}_{625})$
0
0.015
0.035
Confidence Interval
0.55
0.60
0.65
0.70
0.75
0.80
0.85
$\overline{y}_{625}$

The Logic of Confidence Intervals

If the bootstrapped sampling distribution has the same shape as the sampling distribution,

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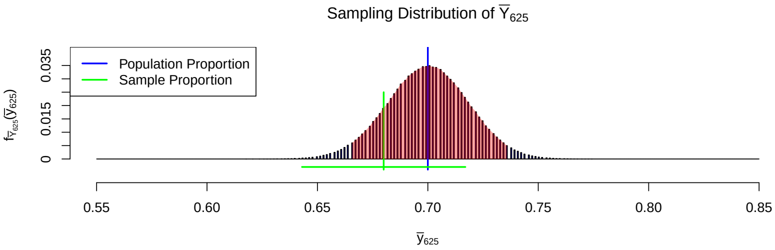
The Logic of Confidence Intervals

If the bootstrapped sampling distribution has the same shape as the sampling distribution,

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.
3. There is a 95% probability of getting a sample proportion in the middle 95% of the sampling distribution, therefore there is 95% probability of getting a 95% confidence interval that covers the population proportion.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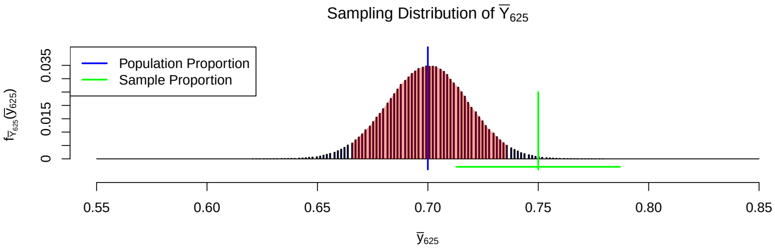
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The Sampling Distribution of the Confidence Interval

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	95% CI
1	1	1	...	0	.68	
2	0	0	...	1	.75	
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
⋮	⋮	⋮	⋮	⋮	⋮	
1 mil	0	1	...	1	.74	
⋮	⋮	⋮	⋮	⋮	⋮	

Sampling Distributions for $\overline{Y}_{625}$ *and the 95% Confidence Interval*

95% of the rows in this sampling distribution will have a CI that covers .70.

The Sampling Distribution of the Confidence Interval

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\overline{Y}_{625}$	95% CI
1	1	1	...	0	.68	[.643,.717]
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⋮	⋮	⋮	⋮	⋮	⋮	

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⋮	⋮	⋮	⋮	⋮	⋮	
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⋮	⋮	⋮	⋮	⋮	⋮	

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3	0	1	...	1	.70	[.663,.737]
4	0	1	...	1	.73	[.694,.766]
5	0	1	...	1	.69	[.653,.727]
⋮	⋮	⋮	⋮	⋮	⋮	⋮
1 mil	0	1	...	1	.74	[.705,.775]
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Sampling Distributions for $\overline{Y}_{625}$ and the 95% Confidence Interval

95% of the rows in this sampling distribution will have a CI that covers .70.

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Informal logic:

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4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1-.7)/625} = .0183$.

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4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1-.7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1-.68)/625} = .0187$.

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2. We can observe that the bootstrapped sampling distribution of the proportion is approximately normally distributed.
3. The variance fully characterizes the shape of a normal distribution (not the location).
4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1-.7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1-.68)/625} = .0187$.
6. When n is large, these variances are close.

Two Questions

1. .68 is close to .70, did we get lucky? No! In a million runs, almost all within .05.
2. Could we have predicted how close we would get before seeing the .70?

Two Questions

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Summary of Population Inference for Proportions